

LECTURE 2

LINEAR REGRESSION MODEL AND OLS

Definitions

A common objective in econometrics is to study the effect of a group of variables, X_i , called the *regressors*, on a variable Y_i , called the *dependent variable*. The econometrician observes data:

$$(Y_1, X_1), (Y_2, X_2), \dots, (Y_n, X_n), \quad (1)$$

where for $i = 1, \dots, n$, Y_i is a random variable and X_i is a random k -vector:

$$X_i = \begin{pmatrix} X_{i1} \\ X_{i2} \\ \vdots \\ X_{ik} \end{pmatrix}.$$

A pair (Y_i, X_i) is called an *observation*, and the collection of observations in (1) is called the *sample*. The vector X_i collects the values of k variables for observation i .

The joint distribution of the sample (1) is called the *population*. The population does not correspond to any physical population; it refers to a probability space. In a *cross-sectional* framework (each observation is a different individual or a firm, etc.), it is often natural to assume that all observations are independently drawn from the same distribution. In this case, the population is described by the distribution of a single observation (Y_1, X_1) . Equivalently, the observations (Y_i, X_i) are iid for $i = 1, \dots, n$. The iid assumption does not imply that Y_i and X_i are independent, but rather that the random vector (Y_i, X_i) is independent of (Y_j, X_j) for $i \neq j$. At the same time, Y_i and X_i can still be related.

In cross-sections, the relationship between the regressors and the dependent variable is modeled through the conditional expectation $E[Y_i | X_i]$. The deviation of Y_i from its conditional expectation is called the *error* (or *disturbance*):

$$U_i = Y_i - E[Y_i | X_i]. \quad (2)$$

Unlike X_i and Y_i , the error U_i is not observable, since the conditional expectation function is unknown to the econometrician.

In the *parametric* framework, it is assumed that the conditional expectation function depends on a number of unknown constants or *parameters*, and that the functional form of $E[Y_i | X_i]$ is known. In the linear regression model, it is assumed that $E[Y_i | X_i]$ is *linear in the parameters*:

$$\begin{aligned} E[Y_i | X_i] &= \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} \\ &= X_i^\top \beta, \end{aligned} \quad (3)$$

where

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix}$$

is a k -vector of unknown constants. The linearity of $E[Y_i | X_i]$ can be justified, for example, by saying that (Y_i, X_i) jointly has a multivariate normal distribution. Since $\beta_j = \partial E[Y_i | X_i] / \partial X_{ij}$, the vector β is a vector of *marginal effects* of X_i , i.e., β_j gives the change in the conditional mean of Y_i per unit change in X_{ij} , while holding the values of other variables (X_{il} for $l \neq j$) fixed. One of the objectives is *estimation* of unknown β from the sample (1).

Combining equations (2) and (3), one can write:

$$Y_i = X_i^\top \beta + U_i. \quad (4)$$

By definition (2),

$$\mathbb{E}[U_i | X_i] = 0.$$

This implies that the regressors contain no information on the deviation of Y_i from its conditional expectation. Further, the law of iterated expectations (LIE) implies that the errors have zero mean: $\mathbb{E}[U_i] = 0$. If (Y_i, X_i) are iid, then the errors $\{U_i : i = 1, \dots, n\}$ are iid as well.

In the *classical regression model*, the variance of the error U_i is assumed to be independent of the regressors and constant across observations:

$$\text{Var}(U_i | X_i) = \sigma^2,$$

for some constant $\sigma^2 > 0$. This property is called *homoskedasticity*.

Assumptions

In this section, we formally define the linear regression model. Define

$$\begin{aligned} X &= \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\ &= \begin{pmatrix} X_{11} & X_{12} & \dots & X_{1k} \\ X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{nk} \end{pmatrix}, \\ Y &= \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \text{ and} \\ U &= \begin{pmatrix} U_1 \\ U_2 \\ \vdots \\ U_n \end{pmatrix}. \end{aligned}$$

The following are the four classical regression assumptions:

(A1) $Y = X\beta + U$.

(A2) $\mathbb{E}[U | X] = 0$ a.s.

(A3) $\text{Var}(U | X) = \sigma^2 I_n$ a.s.

(A4) $\text{rank}(X) = k$ a.s.¹

¹The column (row) rank of a matrix is the maximum number of linearly independent columns (rows). For any matrix, the column and row ranks are equal. If A is an $n \times k$ matrix, then $\text{rank}(A) \leq \min(n, k)$. If $\text{rank}(A) = n$ (or $\text{rank}(A) = k$), we say that A has full row (column) rank. Properties:

$$\begin{aligned} \text{rank}(A) &= \text{rank}(A^\top) = \text{rank}(A^\top A) = \text{rank}(AA^\top), \\ \text{rank}(AB) &\leq \min(\text{rank}(A), \text{rank}(B)), \\ \text{rank}(AB) &= \text{rank}(A) \text{ if } B \text{ is square and of full rank.} \end{aligned}$$

In the classical regression model, instead of conditioning on the regressors, one can assume that X is non-random, i.e., the values of X are fixed in repeated samples. In this case, (A2) becomes $E[U] = 0$ and (A3) becomes $\text{Var}(U) = \sigma^2 I_n$. Since conditioning on X is equivalent to treating it as fixed, the two sets of assumptions lead to the same results.

For inference purposes, sometimes it is assumed that:

$$(A5) \quad U \mid X \sim N(0, \sigma^2 I_n).$$

In the case of fixed regressors, it is assumed instead that the unconditional distribution of the errors is normal. Assumptions (A1)–(A5) define the *classical normal regression model*. In this case,

$$Y \mid X \sim N(X\beta, \sigma^2 I_n).$$

Since all off-diagonal covariance elements in (A5) are zero, (A5) implies independence of the errors. Assumptions (A1)–(A4) alone do not imply independence across observations. For many important results, independence across observations is not required. Nevertheless, sometimes we would like to assume independence without normality:

$$(A6) \quad \{(Y_i, X_i) : i = 1, \dots, n\} \text{ are iid.}$$

In the case of fixed regressors, (A6) can be replaced with the assumption that $\{U_i : i = 1, \dots, n\}$ are iid.

Assumption (A2) says that U is mean independent of X , which is a strong requirement. As noted above, this is equivalent to treating X as non-random and is difficult to justify in many economic applications. However, many important results hold under a weaker condition of uncorrelatedness:

$$(A2^*) \quad \text{For } i = 1, \dots, n, \quad E[U_i X_i] = 0 \text{ and } E[U_i] = 0.$$

Under (A2*), the expression $X_i^\top \beta$ does not have the interpretation of the conditional expectation. In this case, one can treat (4) as a *data-generating process*.

Assumption (A3) implies that the errors U_i have the same variance for all i and are uncorrelated with each other: $E[U_i U_j \mid X] = 0$ for $i \neq j$. If desired, independence of the errors may be achieved through (A5) or Assumptions (A1) and (A6).

Assumption (A4) says that the columns of X are not linearly dependent. If it is violated, then one or more regressors simply duplicate the information contained in other regressors, and thus should be omitted.

Often, the first column of the matrix X is a column of ones. In this case, its coefficient β_1 is called the intercept. The intercept gives the average value of the dependent variable when all regressors are equal to zero.

Estimation by the method of moments

One of the objectives of econometric analysis is *estimation* of unknown parameters β and σ^2 . An *estimator* is any function of the sample $\{(Y_i, X_i) : i = 1, \dots, n\}$. An estimator cannot depend on the unobserved errors U_i or unknown parameters such as β directly; it can depend on unobservables only through the observed variables Y and X . An estimator is not unique in general: several estimators may exist for the same parameter.

One of the oldest methods of finding estimators is called the *method of moments* (MM). The MM replaces population moments (expectations) with the corresponding sample moments (averages). Assumptions (A2) or (A2*) imply that at the true value of β ,

$$\begin{aligned} 0 &= E[U_i X_i] \\ &= E[(Y_i - X_i^\top \beta) X_i]. \end{aligned} \tag{5}$$

Let $\widehat{\beta}$ be an estimator of β . According to the MM, we replace the expectation in (5) with the sample average:

$$\begin{aligned} 0 &= n^{-1} \sum_{i=1}^n (Y_i - X_i^\top \widehat{\beta}) X_i \\ &= n^{-1} \sum_{i=1}^n X_i Y_i - n^{-1} \sum_{i=1}^n X_i X_i^\top \widehat{\beta}. \end{aligned}$$

Here $Y_i - X_i^\top \widehat{\beta}$ is a scalar. Solving for $\widehat{\beta}$, one obtains:

$$\begin{aligned} \widehat{\beta} &= \left(n^{-1} \sum_{i=1}^n X_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_i Y_i \\ &= \left(\sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i \\ &= (X^\top X)^{-1} X^\top Y. \end{aligned} \tag{6}$$

The matrix $\sum_{i=1}^n X_i X_i^\top = X^\top X$ is invertible due to Assumption (A4). To show that $X^\top X = \sum_{i=1}^n X_i X_i^\top$, note that

$$\begin{aligned} X^\top X &= \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix}^\top \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\ &= (X_1 \ X_2 \ \dots \ X_n) \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\ &= X_1 X_1^\top + X_2 X_2^\top + \dots + X_n X_n^\top \\ &= \sum_{i=1}^n X_i X_i^\top. \end{aligned}$$

The expression $X_i^\top \widehat{\beta}$ gives the *estimated regression line*, with $\widehat{Y}_i = X_i^\top \widehat{\beta}$ being the *predicted* value of Y_i , and $\widehat{U}_i = Y_i - X_i^\top \widehat{\beta}$ being the *sample residual*,

$$\widehat{U} = Y - X\widehat{\beta}.$$

The vector $\widehat{U}$ is a function of the estimator of β . For the MM estimator, the sample residuals satisfy the *normal equation*:

$$\begin{aligned} 0 &= X^\top \widehat{U} \\ &= \sum_{i=1}^n \widehat{U}_i X_i \\ &= \begin{pmatrix} \sum_{i=1}^n \widehat{U}_i X_{i1} \\ \sum_{i=1}^n \widehat{U}_i X_{i2} \\ \vdots \\ \sum_{i=1}^n \widehat{U}_i X_{ik} \end{pmatrix}. \end{aligned} \tag{7}$$

If the model contains an intercept, i.e., $X_{i1} = 1$ for all i , then the normal equation implies that $\sum_{i=1}^n \widehat{U}_i = 0$.

To estimate σ^2 , write:

$$\begin{aligned}\sigma^2 &= \mathbb{E}[U_i^2] \\ &= \mathbb{E}\left[(Y_i - X_i^\top \beta)^2\right].\end{aligned}$$

Since β is unknown, we replace it with its MM estimator:

$$\hat{\sigma}^2 = n^{-1} \sum_{i=1}^n \left(Y_i - X_i^\top \hat{\beta}\right)^2.$$

Least Squares

Let b be a candidate value for β . The *ordinary least squares* (OLS) estimator of β minimizes the *sum of squared errors*:

$$\begin{aligned}S(b) &= \sum_{i=1}^n (Y_i - X_i^\top b)^2 \\ &= (Y - Xb)^\top (Y - Xb).\end{aligned}$$

The MM estimator $\hat{\beta}$ coincides with the OLS estimator. To verify this, write

$$\begin{aligned}S(b) &= (Y - Xb)^\top (Y - Xb) \\ &= (Y - X\hat{\beta} + X\hat{\beta} - Xb)^\top (Y - X\hat{\beta} + X\hat{\beta} - Xb) \\ &= (Y - X\hat{\beta})^\top (Y - X\hat{\beta}) \\ &\quad + (X\hat{\beta} - Xb)^\top (X\hat{\beta} - Xb) \\ &\quad + 2(Y - X\hat{\beta})^\top (X\hat{\beta} - Xb) \\ &= (Y - X\hat{\beta})^\top (Y - X\hat{\beta}) \\ &\quad + (\hat{\beta} - b)^\top X^\top X (\hat{\beta} - b) \\ &\quad + 2\hat{U}^\top X (\hat{\beta} - b) \quad (\text{zero by the normal equation}) \\ &= (Y - X\hat{\beta})^\top (Y - X\hat{\beta}) + (\hat{\beta} - b)^\top X^\top X (\hat{\beta} - b).\end{aligned}$$

Minimization of $S(b)$ is equivalent to minimization of $(\hat{\beta} - b)^\top X^\top X (\hat{\beta} - b)$, because $(Y - X\hat{\beta})^\top (Y - X\hat{\beta})$ is not a function of b . If X is of full column rank, as assumed in (A4), $X^\top X$ is a positive definite matrix, and therefore

$$(\hat{\beta} - b)^\top X^\top X (\hat{\beta} - b) \geq 0,$$

where $(\hat{\beta} - b)^\top X^\top X (\hat{\beta} - b) = 0$ if and only if $\hat{\beta} - b = 0$.

Alternatively, one can obtain $\hat{\beta}$ by taking the derivative of $S(b)$ with respect to b and solving the first-order condition $dS(b)/db = 0$. Write

$$S(b) = Y^\top Y - 2b^\top X^\top Y + b^\top X^\top X b.$$

Using the fact that for a symmetric matrix A ,

$$\frac{\partial x^\top A x}{\partial x} = 2Ax,$$

the first-order condition is

$$\left. \frac{\partial S(b)}{\partial b} \right|_{b=\hat{\beta}} = -2X^\top Y + 2X^\top X \hat{\beta} = 0. \quad (8)$$

Solving for $\hat{\beta}$, one obtains:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y. \quad (9)$$

Note also that the first-order condition (8) can be written as $X^\top (Y - X\hat{\beta}) = 0$, which gives us the normal equation (7).

Properties of $\hat{\beta}$

1. $\hat{\beta}$ is a *linear* estimator: an estimator b is called linear if it can be written as $b = AY$, where A is some matrix that depends on X alone and does not depend on Y . In the case of the OLS, $A = (X^\top X)^{-1} X^\top$.
2. Under Assumptions (A1), (A2) and (A4), $\hat{\beta}$ is an *unbiased* estimator, i.e.,

$$E[\hat{\beta}] = \beta.$$

To show unbiasedness, substitute the expression for Y from Assumption (A1) into equation (9):

$$\begin{aligned} \hat{\beta} &= (X^\top X)^{-1} X^\top (X\beta + U) \\ &= \beta + (X^\top X)^{-1} X^\top U. \end{aligned} \quad (10)$$

Next, consider the conditional expectation of $\hat{\beta}$ given X :

$$\begin{aligned} E[\hat{\beta} | X] &= E[\beta + (X^\top X)^{-1} X^\top U | X] \\ &= \beta + E[(X^\top X)^{-1} X^\top U | X]. \end{aligned}$$

Now,

$$\begin{aligned} E[(X^\top X)^{-1} X^\top U | X] &= (X^\top X)^{-1} X^\top E[U | X] \\ &= 0, \end{aligned}$$

since, by Assumption (A2), $E[U | X] = 0$. Therefore,

$$E[\hat{\beta} | X] = \beta. \quad (11)$$

Finally, the LIE implies that

$$\begin{aligned} E[\hat{\beta}] &= E[E[\hat{\beta} | X]] \\ &= \beta. \end{aligned}$$

Equation (11) shows that $\hat{\beta}$ is conditionally unbiased given X . Unbiasedness does not hold under Assumption (A2*) alone.

3. Under Assumptions (A1), (A2) and (A4),

$$\text{Var}(\hat{\beta} | X) = (X^\top X)^{-1} X^\top \text{E}[UU^\top | X] X (X^\top X)^{-1}.$$

Under homoskedastic errors, i.e., Assumption (A3), the expression for the conditional variance simplifies to

$$\text{Var}(\hat{\beta} | X) = \sigma^2 (X^\top X)^{-1}.$$

To show this, we start with the definition of the variance:

$$\begin{aligned} \text{Var}(\hat{\beta} | X) &= \text{E} \left[\left(\hat{\beta} - \text{E}[\hat{\beta} | X] \right) \left(\hat{\beta} - \text{E}[\hat{\beta} | X] \right)^\top | X \right] \\ &= \text{E} \left[\left(\hat{\beta} - \beta \right) \left(\hat{\beta} - \beta \right)^\top | X \right] \\ &= \text{E} \left[(X^\top X)^{-1} X^\top U U^\top X (X^\top X)^{-1} | X \right] \quad (\text{follows from (10)}) \\ &= (X^\top X)^{-1} X^\top \text{E}[UU^\top | X] X (X^\top X)^{-1}. \end{aligned}$$

Under homoskedasticity, $\text{E}[UU^\top | X] = \sigma^2 I_n$, and

$$\begin{aligned} (X^\top X)^{-1} X^\top \text{E}[UU^\top | X] X (X^\top X)^{-1} &= (X^\top X)^{-1} X^\top (\sigma^2 I_n) X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1}. \end{aligned}$$

In the case of fixed regressors, $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$.

4. If we add normality of the errors, i.e., under Assumptions (A1)–(A5), we obtain the following result:

$$\hat{\beta} | X \sim N \left(\beta, \sigma^2 (X^\top X)^{-1} \right).$$

It is sufficient to show that, conditional on X , the distribution of $\hat{\beta}$ is normal. Then, $\hat{\beta} | X \sim N \left(\text{E}[\hat{\beta} | X], \text{Var}(\hat{\beta} | X) \right)$. However, normality of $\hat{\beta} | X$ follows from the facts that $\hat{\beta}$ is a linear function of Y , and that $Y | X$ is normal by Assumption (A5) (see the normal distribution section in Lecture 1). Again, in the case of fixed regressors, we simply omit conditioning on X ,

$$\hat{\beta} \sim N \left(\beta, \sigma^2 (X^\top X)^{-1} \right).$$

5. *Efficiency* (the Gauss–Markov theorem): Under Assumptions (A1)–(A4), the OLS estimator is the Best Linear Unbiased Estimator of β (BLUE), where “best” means having the smallest conditional variance: for any linear and unbiased estimator b , $\text{Var}(b | X) - \text{Var}(\hat{\beta} | X)$ is a positive semidefinite matrix:

$$\text{Var}(b | X) - \text{Var}(\hat{\beta} | X) \geq 0.$$

Furthermore, if $\tilde{\beta}$ is a linear unbiased estimator and $\text{Var}(\tilde{\beta} | X) = \text{Var}(\hat{\beta} | X)$, then $\tilde{\beta} = \hat{\beta}$ with probability one.

Since the theorem concerns conditional variance, the unbiasedness condition refers to conditional unbiasedness: $\text{E}[b | X] = \beta$.

Proof: Let b be a linear and unbiased estimator of β :

$$\begin{aligned} b &= AY, \\ \text{E}[b | X] &= \beta. \end{aligned}$$

These conditions imply that $AX = I_k$ with probability 1. Indeed,

$$\begin{aligned} \mathbb{E}[b \mid X] &= \mathbb{E}[A(X\beta + U) \mid X] \\ &= AX\beta + A\mathbb{E}[U \mid X]. \end{aligned}$$

By Assumption (A2), $\mathbb{E}[U \mid X] = 0$, and thus, for unbiasedness, we need $AX = I_k$. Next, we show that $\text{Cov}(\hat{\beta}, b \mid X) = \text{Var}(\hat{\beta} \mid X)$:

$$\begin{aligned} \text{Cov}(\hat{\beta}, b \mid X) &= \mathbb{E}\left[\left(\hat{\beta} - \beta\right)(b - \beta)^\top \mid X\right] \\ &= \mathbb{E}\left[(X^\top X)^{-1} X^\top U U^\top A^\top \mid X\right] \\ &= (X^\top X)^{-1} X^\top \mathbb{E}[U U^\top \mid X] A^\top \\ &= \sigma^2 (X^\top X)^{-1} X^\top A^\top \text{ (since, by Assumption (A3), } \mathbb{E}[U U^\top \mid X] = \sigma^2 I_n) \\ &= \sigma^2 (X^\top X)^{-1} \text{ (since } X^\top A^\top = I_k) \\ &= \text{Var}(\hat{\beta} \mid X). \end{aligned}$$

Finally,

$$\begin{aligned} \text{Var}(\hat{\beta} - b \mid X) &= \text{Var}(\hat{\beta} \mid X) - \text{Cov}(\hat{\beta}, b \mid X) - \text{Cov}(b, \hat{\beta} \mid X) + \text{Var}(b \mid X) \\ &= \text{Var}(b \mid X) - \text{Var}(\hat{\beta} \mid X). \end{aligned} \tag{12}$$

Since any variance-covariance matrix is positive semidefinite,

$$\text{Var}(b \mid X) - \text{Var}(\hat{\beta} \mid X) \geq 0.$$

To show uniqueness, suppose that there is a linear unbiased estimator $\tilde{\beta}$ such that $\text{Var}(\tilde{\beta} \mid X) = \text{Var}(\hat{\beta} \mid X)$. Then, by (12), $\text{Var}(\hat{\beta} - \tilde{\beta} \mid X) = 0$, and therefore $\tilde{\beta} = \hat{\beta} + c(X)$ for some k -vector-valued function $c(X)$ that can depend only on X . However, since both $\hat{\beta}$ and $\tilde{\beta}$ are conditionally unbiased given X , it follows that $c(X) = 0$ with probability one. $\square$

Note that Assumption (A3), $\mathbb{E}[U U^\top \mid X] = \sigma^2 I_n$, plays a crucial role in the proof of the result. We could not draw conclusions about the efficiency of OLS without it.